

## Killara High School

### Year 11 Term 4 HSC Mathematics Assessment

December 2008

Time Allowed : 60 minutes

Total marks: 46 marks

**Instructions:**

**Read Thoroughly**

All questions should be attempted.

Write your answers neatly, only using **one** side of your answer sheet.

Only **non-programmable** calculators may be used.

Show **all** necessary working.

Marks will be awarded for necessary working.

At the discretion of the examiners, marks may not be awarded for answers **without** adequate working.

At the discretion of the examiners, marks may be deducted for untidy or poorly arranged working.

|        |     |
|--------|-----|
| PART A | /23 |
| PART B | /23 |
| TOTAL  | /46 |

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

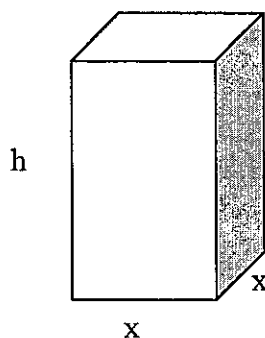
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right) + C$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

**Section A: Geometrical applications of Calculus** (Total marks: 23)

1. A function is defined as  $f(x)$ . The first derivative of  $f(x)$  is defined as  $f'(x)$ . The second derivative of the function is defined as  $f''(x)$ .
- (a) Briefly explain a purpose of finding  $f'(x)$ . (1)
- (b) Rewrite the following, completing the sentence.  
 $f'(x)$  is often referred to as the \_\_\_\_\_ Function. (1)
- (c) If  $f(x)$  is concave up at a point when  $x=a$ , briefly explain what process is used to determine whether the concavity is up or down. (1)
2. A function is defined as  $f(x) = x^3 - 6x^2 + 9x - 5$ .
- (a) Find the Stationary points of the function. (3)
- (b) Find where any points of inflection of the function will lie. (2)
- (c) Determine the nature of each stationary point. (2)
- (d) Draw a neat pair of coordinate axes showing the scale and labeling each axis. (1)
- (e) Plot the stationary points and any points of inflection showing the important features and y-axis intercept. (3)
- (f) The curve will cut the x-axis between two values.  
Find which two values these are. (1)
3. A rectangular block with a square base has the total area of it's surface area equal to  $150 \text{ cm}^2$  and the sides of the base are each  $x \text{ cm}$  long.
- (a) Find an equation for  $h$ , the height of the block in terms of  $x$ . (2)
- (b) Hence show that the volume is  $V = \frac{1}{2}(75x - x^3)$ . (2)
- (c) Hence find the maximum volume of the box. (4)



**Section B: The Quadratic Polynomial.** (Total marks: 23)

1. If  $\alpha$  and  $\beta$  are the roots of the quadratic equation  $5x^2 - x + 15 = 0$ , find without solving the equation the values of:
  - a.  $\alpha + \beta$  (1)
  - b.  $\alpha\beta$  (1)
  - c.  $\alpha^2\beta^3 + \alpha^3\beta^2$  (1)
  - d.  $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$  (2)
  - e.  $\alpha\beta - 2\alpha - 2\beta + 1$  (2)
2. A parabola is defined by the function  $y = x^2 - 2x - 8$ .
  - a. Find the equation of the axis of symmetry. (1)
  - b. Find the co-ordinates of the vertex. (1)
  - c. Sketch the graph of  $y = x^2 - 2x - 8$  showing all the intercepts with the x and y axes. (2)
  - d. Hence solve the inequality:  $x^2 - 2x - 8 < 0$ . (1)
3. Find the value of m for which the quadratic equation  $mx^2 - (2m+1)x + m = 0$  has two real unequal roots. (2)
4. If the equation  $3x^2 + Mx + 8$  has one root which is 6 times the other, find the value of M. (3)
5. For what values of k is the expression  $x^2 - 2kx + k + 2$  positive definite. (3)
6. Find all real roots of the equation  $9^x - 28.3^x + 27 = 0$ . (3)



- Q1 a) • To find the gradient of the tangent to a curve  
• to determine stationary PTS.

(1)

b) Gradient  
or Rabbit

(1)

- c) consider  $f''(a)$  (1)  
if  $f''(a) > 0$  then concave up.  
if  $f''(a) < 0$  then concave down

Q2. a)  $f(x) = x^3 - 6x^2 + 9x - 5$

$$\begin{aligned} f'(x) &= 3x^2 - 12x + 9 \\ &= 3(x^2 - 4x + 3) \\ &= 3(x-1)(x-3) \end{aligned}$$

for st. pts  $f'(x) = 0$

$$(x-1)(x-3) = 0$$

$$x = 1, 3 \quad \checkmark$$

$(1, -1)$  and  $(3, -5)$  ✓

(3)

b)  $f''(x) = 6x - 12$   
 $= 6(x-2)$

| x        | 1 | 2 | 3 |
|----------|---|---|---|
| $f''(x)$ | - | 0 | + |

change of concavity ✓

$$(x-2)=0 \Rightarrow x=2 \quad \checkmark$$

c)  $f''(1) = 6 - 12 = -6 < 0$

✓  
∴ max at  $(1, -1)$

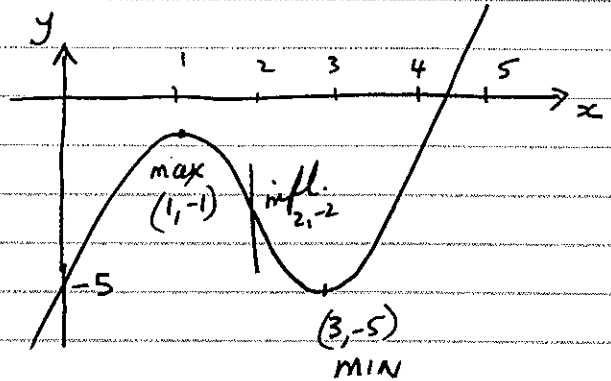
$$f''(3) = 18 - 12 = 6 > 0$$

✓  
∴ min at  $(3, -5)$

(2)

$(1, -1)$   
 $(3, -5)$

$-5$   $3, -5$



- d) • neat axes / labelled ✓  
• (scale) units ✓

(1)

e)

$$f(4) = -1$$

$$f(5) = 15$$

graph as above  
plot + label

- inflection ✓
- max ✓
- min ✓
- y intercept ✓
- approx x intercept ✓

-1 for each missing

(3)

(f) Between 4 and 5 (1)

Q3. a)

$$\text{Surface area} = 2x^2 + 4xh \quad \checkmark$$

$$\therefore 150 = 2x^2 + 4xh$$

$$h = \frac{150 - 2x^2}{4x} \quad \checkmark$$

(2)



$$b) V = Ah$$

$$= x^2 h$$

$$= \left( \frac{150 - 2x^2}{4x} \right) \cdot x^2 \quad \checkmark \text{ce}$$

$$= \frac{x}{4} (150 - 2x^2)$$

$$V = \frac{x}{2} (75 - x^2) \quad \checkmark \quad (2)$$

$$c) V = \frac{75x}{2} - \frac{x^3}{2}$$

$$\frac{dV}{dx} = \frac{75}{2} - \frac{3x^2}{2} \quad \checkmark$$

$$\text{for max/min } \frac{dV}{dx} = 0$$

$$\frac{3x^2}{2} = \frac{75}{2}$$

$$x^2 = 25$$

$$x = \pm 5$$

$$x = 5 \quad \text{but } x > 0$$

$$x = 5 \quad \checkmark$$

$$\frac{d^2V}{dx^2} = -\frac{6x}{2} = -3x$$

$$\text{if } x = 5 \quad \frac{d^2V}{dx^2} = -15 < 0$$

$$\therefore \text{max vol when } x = 5 \quad \checkmark \quad (4)$$

$$\text{max vol} = \frac{5}{2} (75 - 25)$$

$$= \frac{5(50)}{2}$$

$$= \frac{250}{2}$$

$$= 125 \text{ cm}^3 \quad \checkmark$$

### Section B

Q4.

$$5x^2 - x + 15 = 0$$

$$a) \alpha + \beta = -\frac{(-1)}{5} = \frac{1}{5} \quad \checkmark \quad (1)$$

$$b) \alpha \beta = \frac{15}{5} = 3 \quad \checkmark \quad (1)$$

$$c) \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2 \beta^2} \quad \checkmark$$

$$= \frac{\left(\frac{1}{5}\right)^2 - 2(3)}{(3)^2}$$

$$= \frac{\frac{1}{25} - 6}{9}$$

$$= \frac{-149}{225} \quad \checkmark \quad (2)$$

d)

$$\alpha\beta - 2\alpha - 2\beta + 1$$

$$= \alpha\beta - 2(\alpha + \beta) + 1$$

$$= 3 - 2\left(\frac{1}{5}\right) + 1$$

$$= 3 - \frac{2}{5} + 1$$

$$= \frac{18}{5}$$

$$e) \alpha^2 \beta^3 + \alpha^3 \beta^2$$

$$= \alpha^2 \beta^3 (\beta + \alpha)$$

$$= 3^2 \left(\frac{1}{5}\right) = \frac{9}{5} \quad \checkmark \quad (1)$$

Q5

$$y = x^2 - 2x - 8$$

a) axis of symmetry  $x = -\frac{b}{2a}$

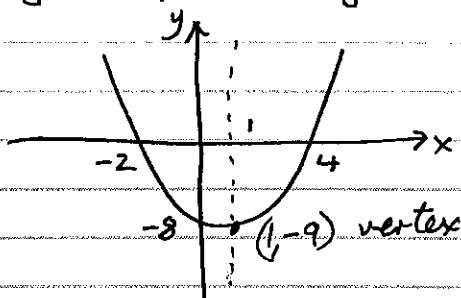
$$x = -\frac{(-2)}{2(1)} = 1 \quad \checkmark \quad (1)$$

b) at vertex  $x=1$

$$y = 1 - 2 - 8 = -9$$

$$(1, -9) \quad \checkmark \quad (1)$$

c) if  $y=0$   $x^2 - 2x - 8 = 0$   
 $(x-4)(x+2) = 0$

x intercepts  $x = 4, -2 \quad \checkmark$   
for y intercept  $x=0$ ,  $y = -8 \quad \checkmark$ 

d)  $x^2 - 2x - 8 < 0$

$$-2 < x < 4 \quad \checkmark \quad (1)$$

Q6  $mx^2 - (2m+1)x + m = 0$

for 2 real roots  $\Delta > 0$   
distinct

$$b^2 - 4ac = [-(2m+1)]^2 - 4(m)(m) > 0 \quad \checkmark \text{ ex } \&$$

$$= (2m+1)^2 - 4m^2$$

$$= 4m^2 + 4m + 1 - 4m^2$$

$$= 4m + 1 > 0$$

$$4m > -1$$

$$m > -\frac{1}{4}$$

$$\checkmark \quad (2)$$

Q7 Let roots be  
 $p, 6p$ 

$$p + 6p = -\frac{M}{3}$$

$$7p = -\frac{M}{3} \quad M = -21p \quad (A) \quad \checkmark$$

$$p \cdot 6p = \frac{8}{3}$$

$$6p^2 = \frac{8}{3}$$

$$p^2 = \frac{8}{18} = \frac{4}{9}$$

$$p = \pm \frac{2}{3} \quad (B) \quad \checkmark$$

Sub (B) into (A)

$$M = -21 \left( \pm \frac{2}{3} \right)$$

$$M = \pm 14 \quad \checkmark \quad (3)$$

Q8

positive definite

$$\Delta < 0 \quad a > 0$$

$$a = 1 > 0$$

$$\Delta = b^2 - 4ac$$

$$= (-2k)^2 - 4(1)(k+2)$$

$$= 4k^2 - 4k - 8$$

$$= 4(k^2 - k - 2)$$

$$= 4(k-2)(k+1)$$

$$4(k-2)(k+1) < 0 \quad \checkmark$$

$$-1 < x < 2 \quad \checkmark \quad (3)$$



Question 9.

$$9^x - 28 \cdot 3^x + 27 = 0$$

$$\text{Let } m = 3^x \quad \therefore m^2 = (3^x)^2 = (3^x)^x = 9^x$$

$$m^2 - 28m + 27 = 0$$

$$(m-27)(m-1) = 0$$

$$m = 27 \text{ or } 1.$$

✓

$$3^x = 27 \text{ or } 3^x = 1$$

$$x = 3 \text{ or } x = 1$$

✓✓

(3)

~~23~~